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Task: PEAS Description and Task Environment Properties of Intelligent Agents

Introduction

In Artificial Intelligence (AI), an intelligent agent perceives its environment through sensors and acts upon it using actuators to achieve a specific goal. To describe an agent's task environment, we use the PEAS framework and classify the environment based on its properties.

PEAS stands for:

- P – Performance Measure: The criteria used to evaluate the agent's success.
- E – Environment: The surroundings in which the agent operates.
- A – Actuators: The tools or mechanisms used by the agent to perform actions.
- S – Sensors: The tools used by the agent to perceive information from the environment.

The main task environment properties are:

1. Fully observable / Partially observable: Whether the agent can access all relevant information about the environment.
2. Deterministic / Stochastic: Whether an action produces a predictable result or involves uncertainty.
3. Episodic / Sequential: Whether each decision is independent or affects future decisions.
4. Static / Dynamic: Whether the environment changes while the agent is making decisions.

5. Discrete / Continuous: Whether states, actions, or time are represented as separate values or vary continuously.
6. Single-agent / Multi-agent: Whether one agent operates alone or interacts with other agents.
7. Known / Unknown: Whether the agent knows the rules and effects of its actions.

A) Playing Soccer PEAS Description

- Performance Measure: Scoring goals, preventing the opponent from scoring, winning the match, teamwork, and following the rules.
- Environment: Soccer field, ball, goalposts, teammates, opponents, referee, and match conditions.
- Actuators: Legs and feet for running, kicking, and passing; head for heading the ball; and body movements for defending.
- Sensors: Eyes for observing the ball and players, ears for hearing whistles and instructions, and body sensors for balance and movement.

Task Environment Properties

Property	**Classification**	**Explanation**
Observability	Partially observable	A player cannot see every player and event simultaneously, especially those outside their field of vision.
Determinism	Stochastic	The outcome of a kick or pass depends on opponents, ball movement, and other unpredictable factors.

Task structure	Sequential	Every action, such as passing or shooting, influences subsequent actions and the final result.
Dynamics	Dynamic	Players and the ball move continuously while decisions are being made.
State and actions	Continuous	Player positions, speed, direction, and ball movement vary continuously, although the game also has discrete events such as goals.

B) Exploring the Subsurface Oceans of Titan

Titan is Saturn's largest moon. Exploring its subsurface ocean involves using autonomous robotic systems or probes to investigate an unfamiliar environment beneath its surface.

PEAS Description

- Performance Measure: Successful exploration, scientific data collection, accurate mapping, discovery of chemical compounds, energy efficiency, and safe operation.
- Environment: Titan's surface, icy crust, subsurface ocean, underwater terrain, possible obstacles, and extreme environmental conditions.
- Actuators: Robotic propulsion systems, steering mechanisms, robotic arms, sampling devices, and communication equipment.
- Sensors: Cameras, sonar, pressure sensors, temperature sensors, chemical analyzers, navigation instruments, and scientific measurement devices.

Task Environment Properties

Property	Classification	Explanation
Observability	Partially observable	The probe cannot directly observe the entire ocean and must use sensors to discover hidden features.
Determinism	Stochastic	Uncertain terrain, equipment behavior, and environmental conditions can affect exploration.
Task structure	Sequential	Each movement, measurement, and route choice affects later exploration activities.
Dynamics	Dynamic	Currents and local environmental conditions may change, although some parts of the environment may remain relatively stable.
State and actions	Continuous	Position, depth, pressure, temperature, and movement vary continuously.

C) Shopping for Used AI Books on the Internet PEAS Description

- Performance Measure: Finding the required AI books, obtaining accurate editions, minimizing total cost, checking book condition, ensuring seller reliability, and receiving the books successfully.
- Environment: Online bookstores, second-hand marketplaces, seller listings, customer reviews, payment systems, and delivery services.
- Actuators: Keyboard and mouse inputs, search queries, filters, clicks, comparison tools, shopping cart actions, and checkout operations.

- Sensors: Webpage content, book descriptions, prices, seller ratings, photographs, availability information, delivery estimates, and customer reviews.

Task Environment Properties

Property	**Classification**	**Explanation**
Observability	Partially observable	Listings provide information, but the actual book condition and seller reliability may not be fully known.
Determinism	Stochastic	Prices, stock availability, seller responses, and delivery outcomes may change unpredictably.
Task structure	Sequential	Searching, comparing, selecting, purchasing, and tracking delivery are connected steps.
Dynamics	Dynamic	Prices, listings, and availability can change while the agent is shopping.
State and actions	Mostly discrete, with continuous-valued attributes	Actions such as clicking and selecting are discrete, while prices and delivery times are numerical values.

D) Playing a Tennis Match PEAS Description

- Performance Measure: Winning points and games, winning the match, making accurate shots, minimizing unforced errors, and maintaining effective court positioning.

- Environment: Tennis court, net, ball, opponent, umpire, court boundaries, and weather conditions for outdoor matches.
- Actuators: Arms, hands, legs, and body movements used for serving, hitting the ball, running, and changing position.
- Sensors: Vision for tracking the ball and opponent, hearing for instructions and calls, and proprioceptive sensors for balance and body position.

Task Environment Properties

Property	Classification	Explanation
Observability	Partially observable	The player may not perfectly predict the opponent's intentions, spin, or next shot.
Determinism	Stochastic	Shot outcomes depend on timing, ball spin, opponent responses, and environmental conditions.
Task structure	Sequential	Each serve and return affects the next shot and the overall point.
Dynamics	Dynamic	The ball and players move rapidly during play.
State and actions	Continuous	Ball position, velocity, racket angle, and player movement vary continuously.

E) Practicing Tennis Against a Wall PEAS Description

- Performance Measure: Improving accuracy, consistency, reaction time, shot control, footwork, and the number of successful consecutive returns.
- Environment: Practice area, wall, tennis ball, racket, court markings, and surrounding obstacles.
- Actuators: Arms and hands for racket control, and legs and body movements for positioning and returning the ball.
- Sensors: Vision for tracking the ball, proprioception for body position, and auditory feedback for identifying ball contact and bounce timing.

Task Environment Properties

Property	Classification	Explanation
Observability	Mostly fully observable	The ball and wall are generally visible, although exact spin and contact details may be difficult to estimate.
Determinism	Approximately deterministic	The ball's rebound is relatively predictable under stable conditions, but spin and imperfect contact introduce uncertainty.
Task structure	Sequential	Every shot determines the ball's next rebound and the player's next movement.
Dynamics	Dynamic	The ball moves continuously, requiring immediate responses.
State and actions	Continuous	Ball position, speed, racket angle, and player movements vary continuously.

F) Performing a High Jump PEAS Description

- Performance Measure: Achieving the maximum valid height, clearing the bar without knocking it down, completing the attempt within the rules, and avoiding injury.
- Environment: Running track, take-off area, crossbar, uprights, landing mat, and competition conditions.
- Actuators: Legs, arms, torso, and body movements used for running, take-off, rotation, and landing.

Task Environment Properties

- Sensors: Vision for judging the bar and take-off point, proprioception for body position, and balance and movement sensors for coordinating the jump.

Property	**Classification**	**Explanation**
Observability	Mostly fully observable	The bar, runway, and landing area are visible, although internal body forces and exact movement states may not be directly observable.
Determinism	Approximately deterministic	The jump follows physical laws, but small variations in technique, friction, and body movement affect the result.
Task structure	Episodic	Each attempt is a separate trial, although previous attempts can influence later strategy.
Dynamics	Dynamic	The athlete's body moves rapidly throughout the jump.
State and actions	Continuous	Speed, force, body angle, trajectory, and jump height vary continuously.
Agents	Single-agent	The athlete performs the jump independently, although competitors and officials are present.

Task Environment Properties

G) Knitting a Sweater PEAS Description

- Performance Measure: Producing a correctly sized sweater, maintaining stitch accuracy, achieving the required pattern and quality, minimizing material waste, and completing the work efficiently.
- Environment: Yarn, knitting needles or a knitting machine, the partially completed sweater, pattern instructions, and working area.

- Actuators: Hands and fingers controlling the needles and yarn, or motors and mechanical components in an automated knitting machine.
- Sensors: Vision for checking stitches and patterns, touch sensors for yarn tension, and position sensors for monitoring needle movement in an automated system.

Task Environment Properties

Property	**Classification**	**Explanation**
Observability	Mostly fully observable	The visible stitches and pattern can usually be inspected, although hidden defects and yarn tension may be difficult to detect.
Determinism	Mostly deterministic	Following the same knitting instructions generally produces predictable results, but yarn behavior and human errors introduce uncertainty.
Task structure	Sequential	Each stitch depends on previous stitches and contributes to the final garment.
Dynamics	Mostly static, with dynamic aspects	The workpiece remains relatively stable between actions, but its shape changes as knitting progresses.
State and actions	Mostly discrete actions with continuous variables	Individual stitches are discrete actions, while yarn tension and fabric dimensions vary continuously.

Agents	Single-agent	One person or knitting machine performs the task, even though another person may provide
		instructions.

H) Bidding on an Item at an Auction PEAS Description

- Performance Measure: Obtaining the desired item at a favorable price, staying within the budget, maximizing value, and completing the purchase successfully.
- Environment: Auction platform or auction hall, item listing, seller, competing bidders, auctioneer, bidding rules, and payment system.
- Actuators: Submitting bids, raising a bidding paddle, entering a maximum bid, placing an online bid, and confirming payment when authorized.
- Sensors: Current highest bid, bid history, auction countdown, item description, seller information, competing bids, and auction rules.

Task Environment Properties

Property	**Classification**	**Explanation**
Observability	Partially observable	The agent may see the current highest bid but cannot know every competitor's maximum budget or bidding strategy.
Determinism	Stochastic	The outcome depends on competing bids, timing, and other bidders' decisions.

Task structure	Sequential	Each bid affects the agent's future options, remaining budget, and chance of winning.
Dynamics	Dynamic	Competing bidders may submit new bids, and the auction state changes over time.

Overall Comparison of the Eight Task Environments

Activity	**Observability**	**Determinism**	**Task Structure**	**Dynamics**	**Agents**
A) Playing soccer	Partially observable	Stochastic	Sequential	Dynamic	Multi agent
B) Exploring Titan's subsurface oceans	Partially observable	Stochastic	Sequential	Dynamic	Usually singl eagent
C) Shopping for used AI books	Partially observable	Stochastic	Sequential	Dynamic	Multi agent
D) Playing a tennis match	Partially observable	Stochastic	Sequential	Dynamic	Multi agent

Conclusion

The PEAS framework helps define an intelligent agent's objectives, surroundings, available actions, and sources of information. Activities such as soccer, tennis, and auctions require agents to respond to competing agents and unpredictable events. Exploring Titan requires autonomous decisions under limited knowledge, while online shopping involves comparing alternatives and managing uncertainty. High jumping is a short, performance-focused task, and knitting requires careful sequential execution. Understanding these properties helps AI designers select suitable sensing methods, decision-making strategies, and learning techniques for each task environment.